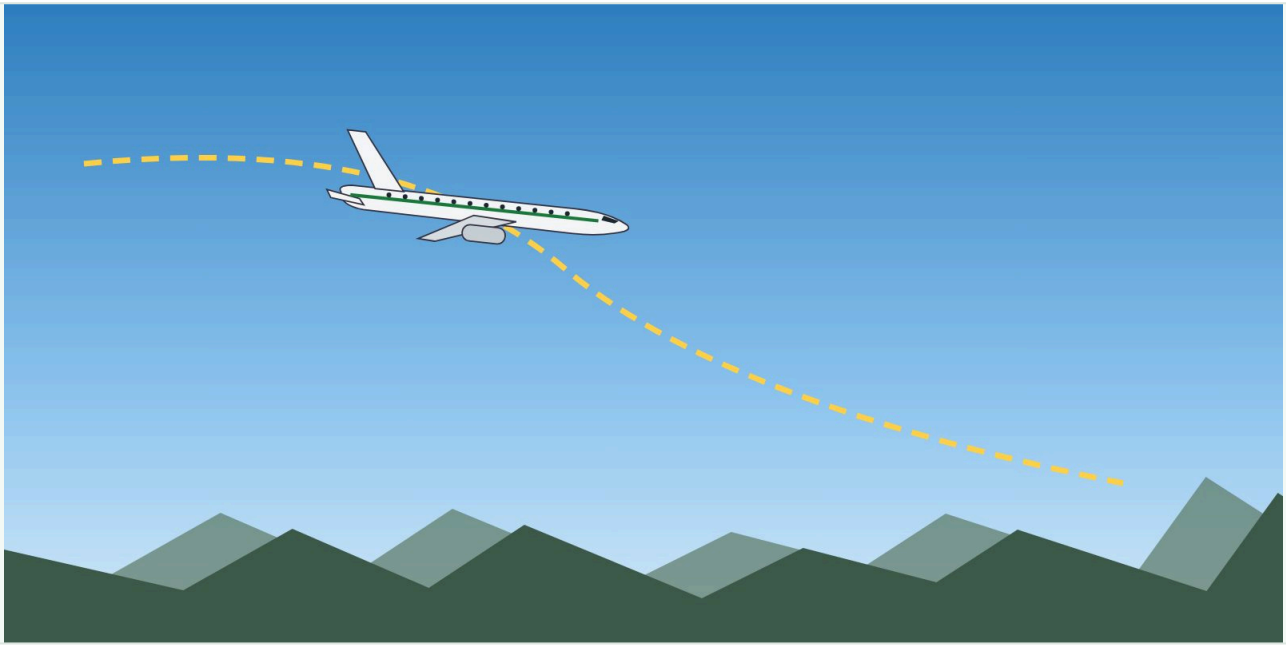
MODULE 1 · PBN PRINCIPLES

Lesson 01: The PBN Concept

Lesson 1 of 5 · about 7 min

LEARNING OBJECTIVES

- Explain the performance-based navigation (PBN) concept.
- Distinguish RNAV and RNP specifications.
- Identify the elements of a PBN operation.



PBN: the performance matters, not the sensor.

From sensors to performance

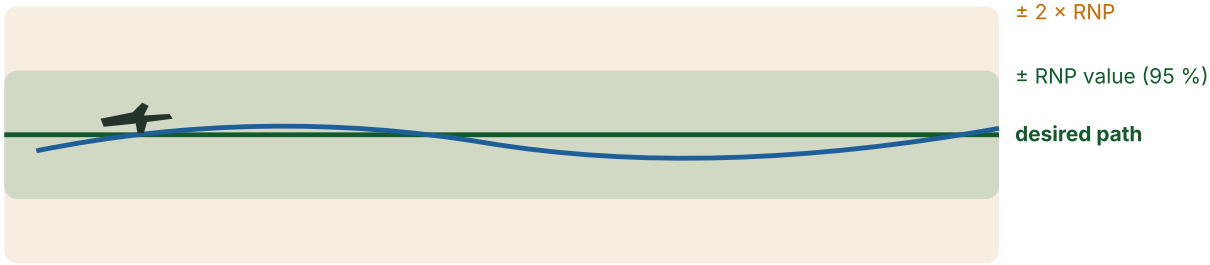
Conventional navigation is based on flying to and from ground beacons (VOR, NDB). **Performance-based navigation (PBN)** (ICAO Doc 9613) defines instead the **performance** an aircraft's area navigation system must achieve — accuracy, integrity, continuity and functionality — for a given airspace or procedure.

A PBN operation has three elements:

- the **navigation specification** — the aircraft, system and crew requirements;
- the **navaid infrastructure** — GNSS, DME/DME, VOR/DME;
- the **navigation application** — the routes and procedures designed for it.

RNAV and RNP

RNP: the aircraft must stay within ± the RNP value at least 95 % of the flight time —
and an RNP system must monitor its own performance and alert the crew when it cannot.



actual flight path = path definition error + flight technical error + navigation system error

Total system error combines the path definition, flight technical and navigation system errors. On-board performance monitoring and alerting is what makes a specification “RNP”.

- An **RNAV** specification has no requirement for on-board performance monitoring and alerting.
- An **RNP** specification **requires on-board performance monitoring and alerting** — the system tells the crew when it cannot meet the required performance (for example “UNABLE REQD NAV PERF – RNP” or a similar alert).
- The number (e.g. RNP 1) is the lateral accuracy in nautical miles that must be achieved at least 95 % of the flight time.

Navigation specifications

Phase	Typical specifications
Oceanic / remote	RNAV 10, RNP 4, RNP 2
En route continental	RNAV 5 (B-RNAV), RNAV 2, RNP 2, Advanced RNP
Terminal (SID / STAR)	RNAV 1, RNP 1, Advanced RNP
Approach	RNP APCH (LNAV, LNAV/VNAV, LP, LPV), RNP AR APCH

The aircraft's **AFM / AFM supplement** states which specifications it is certified for, and the **MEL** gives the dispatch requirements. Check your fleet's capabilities in the OM B.

KEY TAKEAWAYS

- PBN = navigation specification + navaid infrastructure + navigation application.
- RNP requires on-board performance monitoring and alerting; RNAV does not.
- RNP 1 = ±1 NM at least 95 % of the time.
- Fleet capability: AFM, OM B and MEL.

KNOWLEDGE CHECK

What distinguishes an RNP specification from an RNAV specification?

A. A smaller number

B. On-board performance monitoring and alerting

C. The use of VOR

D. Autopilot coupling

Answer: B. RNP requires on-board performance monitoring and alerting.

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.